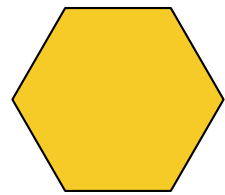
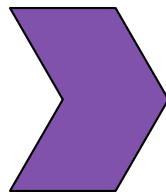
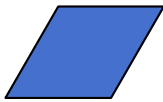
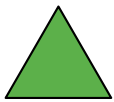
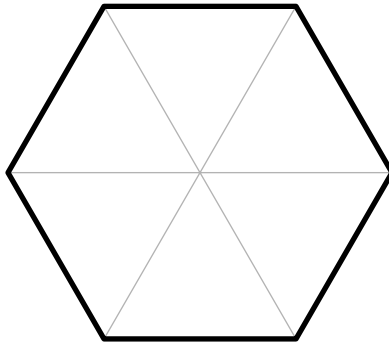
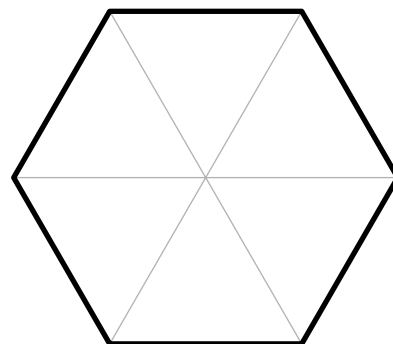
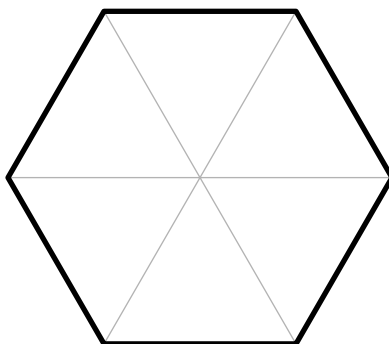
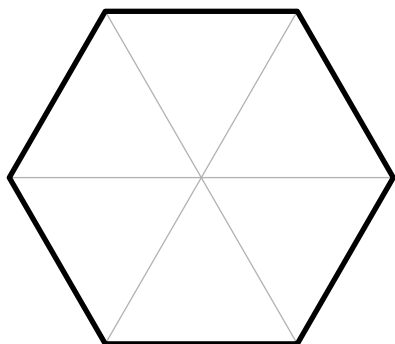
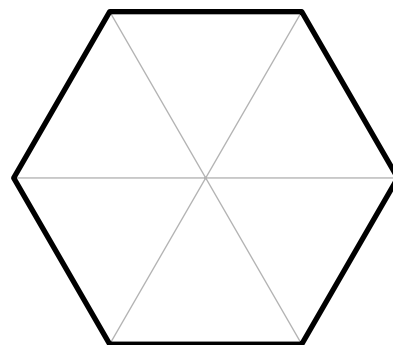
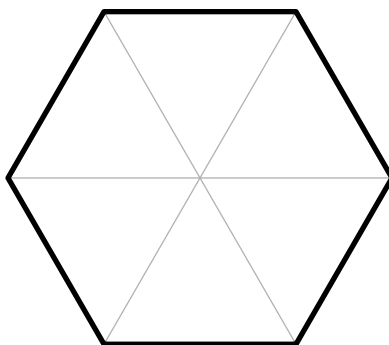
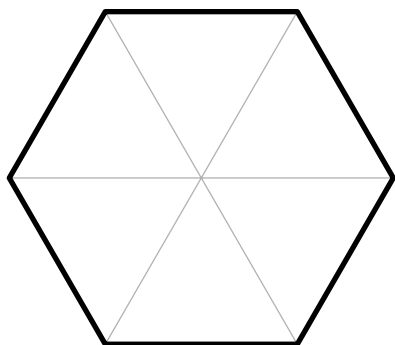
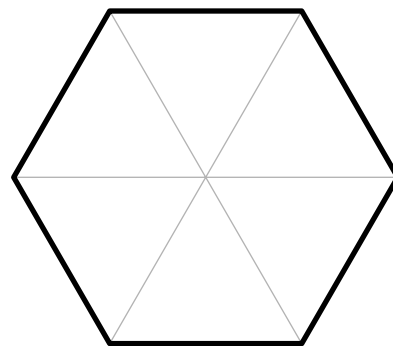
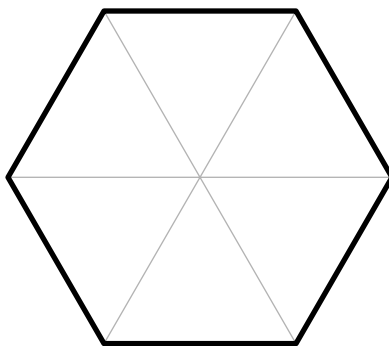
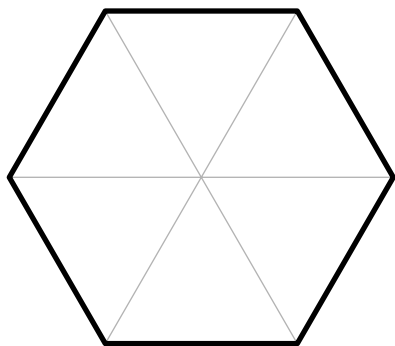
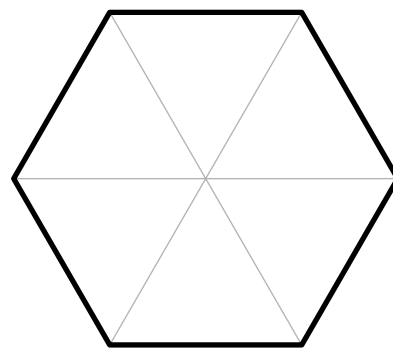
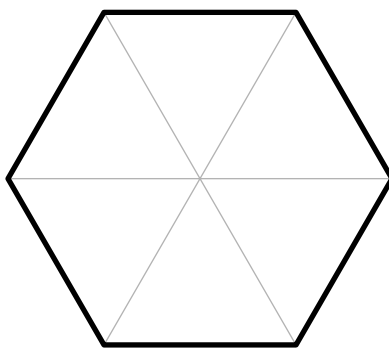
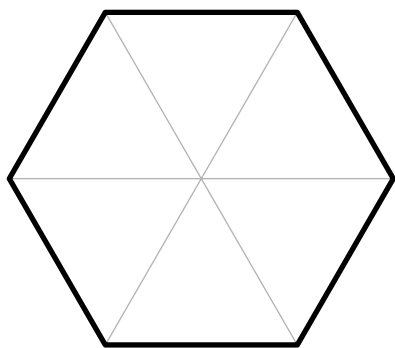


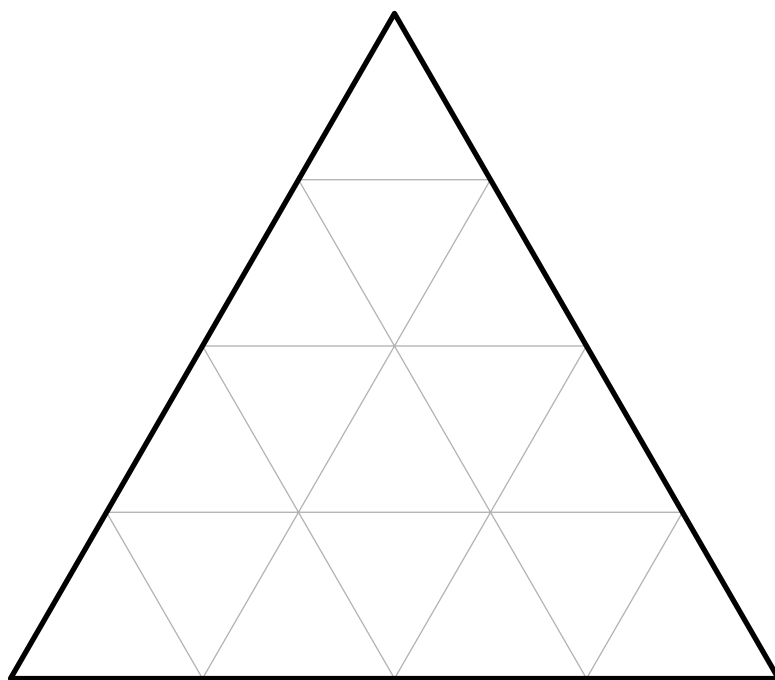
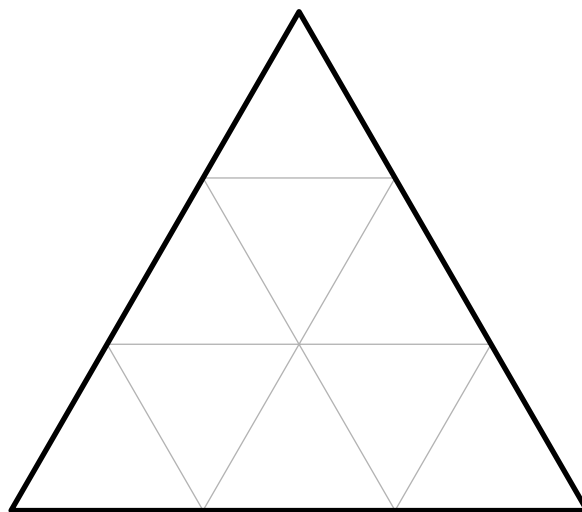
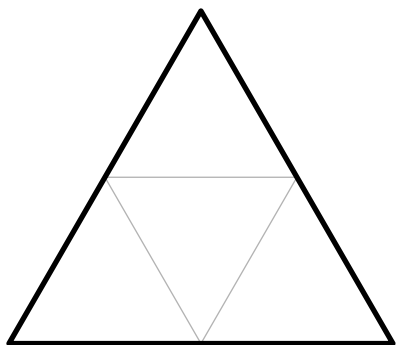
Problem 1: Fill the hexagon with pieces that are all the same color. Which colors can do it? Under each piece, write how many pieces of that color fill the hexagon.



Problem 2: Find different ways to fill the hexagon. Two ways are the same if they use the same pieces. Draw each new way in one of the hexagons below. How many different ways are there?

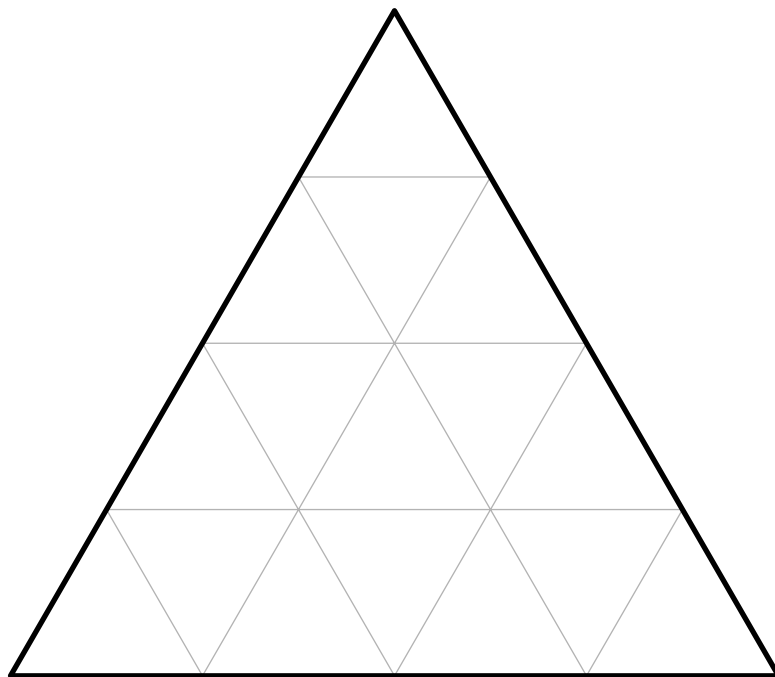
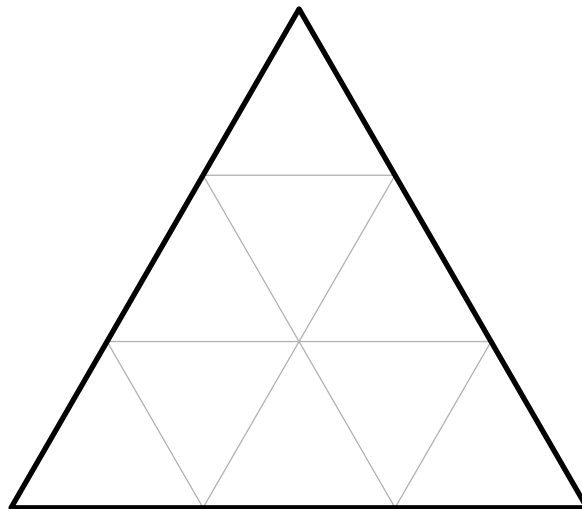
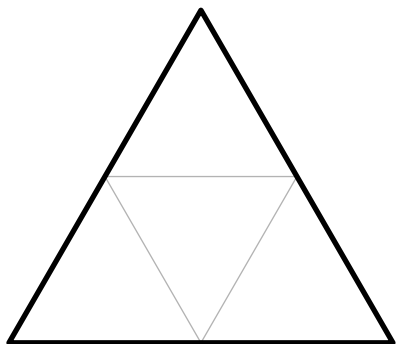


Problem 3: Fill each big triangle with green triangles. Write how many green triangles it takes.



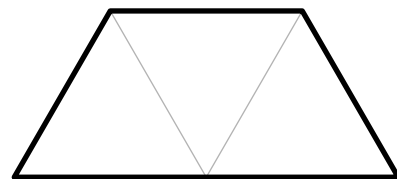
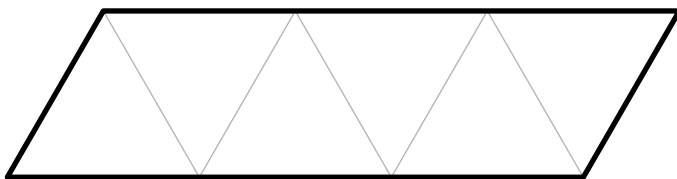
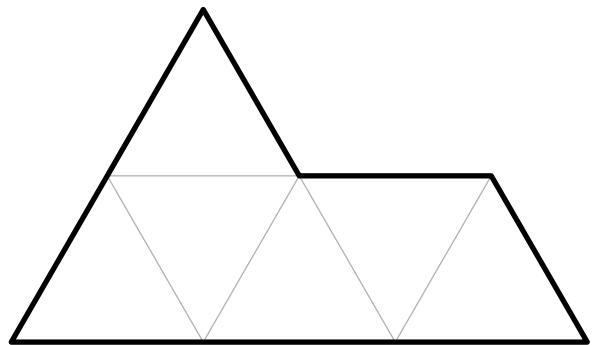
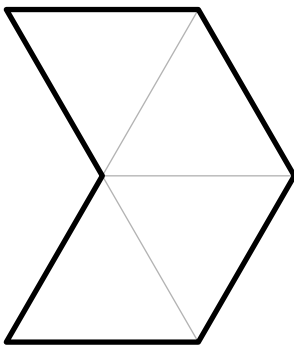
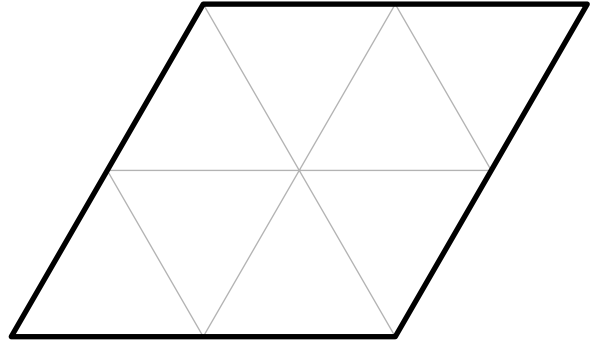
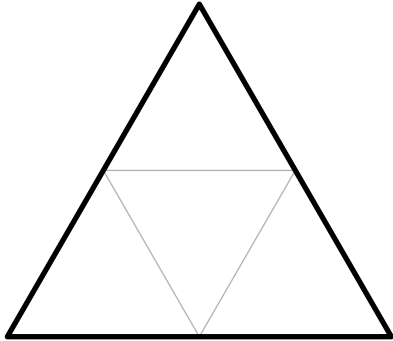
How many green triangles would fill a big triangle that is 5 green triangles long on each side?

Problem 4: Fill each big triangle with blue pieces and green triangles. Use as few green triangles as you can. Write how many green triangles you used.

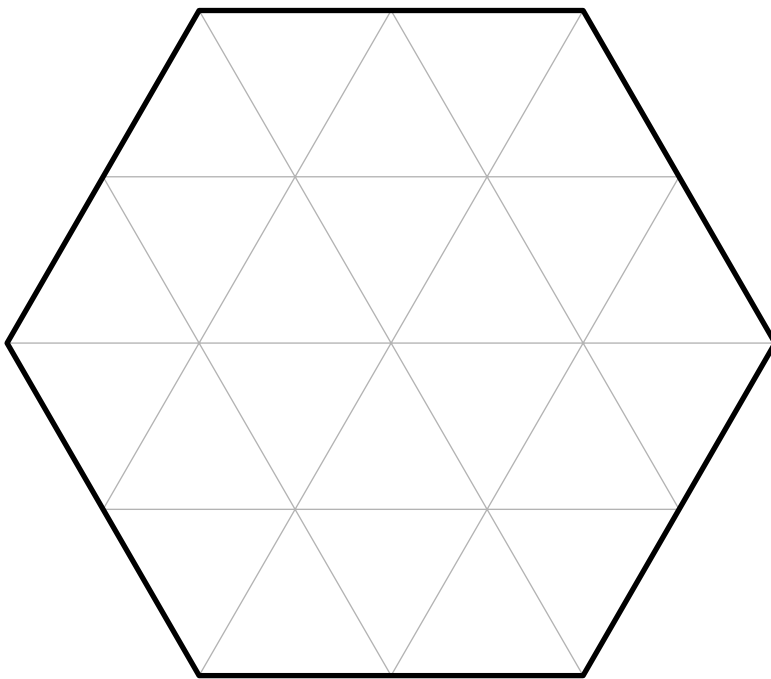
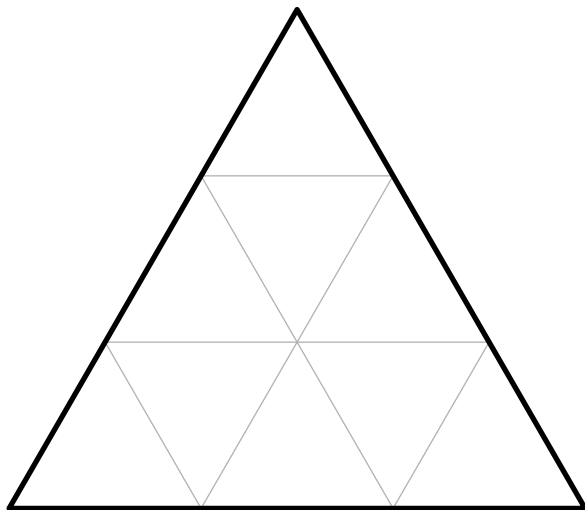


How few green triangles would a big triangle that is 5 green triangles long on each side need?

Problem 5: Circle each shape that you can fill with blue pieces only. For a shape you cannot fill, how do you know?



Problem 6: Fill the big triangle with as few pieces as you can. You may use any pieces. Then do the same for the big hexagon. Write how many pieces you used.



Problem 7: Put 6 green triangles together, edge to edge, to make a shape that blue pieces cannot fill. Draw your shape on the grid. Then make a shape from 8 green triangles that blue pieces cannot fill.

